

OTIF ROOT-CAUSE ENGINE

CASE STUDY Attributing delivery failures to a root cause, with the money next to each one.

95.7%

OVERALL ACCURACY
canonical batch, seed 46

85.5%

MEAN ON HARD ORDERS
across 6 independent batches

+23.7

POINTS MEAN LIFT
vs a simple biggest-number rule

AED 1,226,642

FAILURE VALUE ATTRIBUTED
ranked by cash, seed 46

The problem

When on-time-in-full slips, nobody agrees on why. Planning blames the suppliers, the suppliers blame the forecast, the warehouse blames both, and logistics says the order left on time. It costs days of meetings and finger-pointing, and no single owner ever agrees to hold the number. The cause is real and knowable, but it is buried in per-order signals no one has time to reconcile by hand.

What the tool does

Four specialist agents (one each for supplier, demand, warehouse, and logistics) investigate every failed order in parallel. A coordinator reads all four findings and names the one dominant cause, then ranks the causes by the cash behind them. The agents reason; they never produce a figure. Every number in the output is computed by tested code and checked before it can be shown, so nothing the model writes can invent a value.

Where the value sits (canonical batch, seed 46)

DRIVER	VALUE AT STAKE	SHARE	FAILED ORDERS	RECOMMENDED ACTION
Supplier	AED 477,794	39.0%	51	Qualify a second source for the worst-offending vendors
Warehouse	AED 313,864	25.6%	37	Clear the dispatch bottleneck with a hard release cut-off
Demand	AED 282,823	23.1%	28	Recalibrate the forecast on the SKUs that consistently under-call
Logistics	AED 152,160	12.4%	24	Review and re-bond the chronically overrun lanes

Total failure value across all 140 failed orders in this batch: AED 1,226,642. Shares are of cash at risk and are rounded independently.

How it works: four specialists, one coordinator

Prep is deterministic: the code reads each failed order and computes the raw signals (late days, shortfall, dispatch slip, transit overage). Four specialist agents then run in parallel on Sonnet 4.6, each seeing only its own domain's evidence and never the answer key. The coordinator adjudicates the orders where more than one cause fired, choosing the deciding one rather than the biggest number. Finalize is deterministic again: it rolls up the cash, ranks the drivers, and renders the brief through a gate that blocks any number the code did not produce. The LLM reasons; tested code computes.

The robustness story

One good run proves nothing. Accuracy here is measured across six independent batches (seeds 42 to 47), each generated fresh with different planted causes the engine has never seen. Mean accuracy on the hard orders, the ones where two teams are genuinely at fault at once, is 85.5% (range 78.1% to 93.9%), against 61% to 84% for a simple biggest-number rule. On one batch the simple rule actually won. We show that row too.

BATCH (SEED)	HARD-ORDER ACCURACY	SIMPLE RULE	LIFT	NOTE
42	78.1%	53.1%	+25.0	
43	93.9%	60.6%	+33.3	
44	89.2%	56.8%	+32.4	
45	90.3%	54.8%	+35.5	
46	83.3%	61.1%	+22.2	canonical, shown on page 1
47	78.1%	84.4%	-6.2	simple rule won

Mean lift across the six batches: +23.7 points on the hard orders. Mean overall accuracy: 96.5%.

Why the numbers can be trusted

- 1. No fabrication.** A firewall keeps the planted ground truth away from the agents, and a render gate blocks any digit the model authored from reaching the brief. Only code-computed, placeholder-bound numbers get through.
- 2. Measured, not asserted.** Accuracy is scored against the cause planted in each order when the batch was generated, not against the model's own say-so.
- 3. Honest about the loss.** On seed 47 the simple rule beat the coordinator. That batch stays in the table at full weight rather than being quietly dropped.

Worked example: order OTIF-0011

The supplier ran 10 days late; the warehouse was 2 days slow. The simple rule picks the biggest single signal and blames the warehouse. Wrong.

The coordinator reasons that the late PO alone would have closed the shortfall had it arrived on time, so it names the supplier as the deciding cause. Correct, and it shows the reasoning.

Built with Python, FastAPI, LangGraph, Claude Sonnet 4.6, Next.js. Verified by 181 passing tests.

Mawarid Distribution is a fictional company and all order data is synthetic, generated with a fixed seed so every figure in this document is reproducible. Prepared by Prajwal.